STATA Tutorial 2A

Data Management

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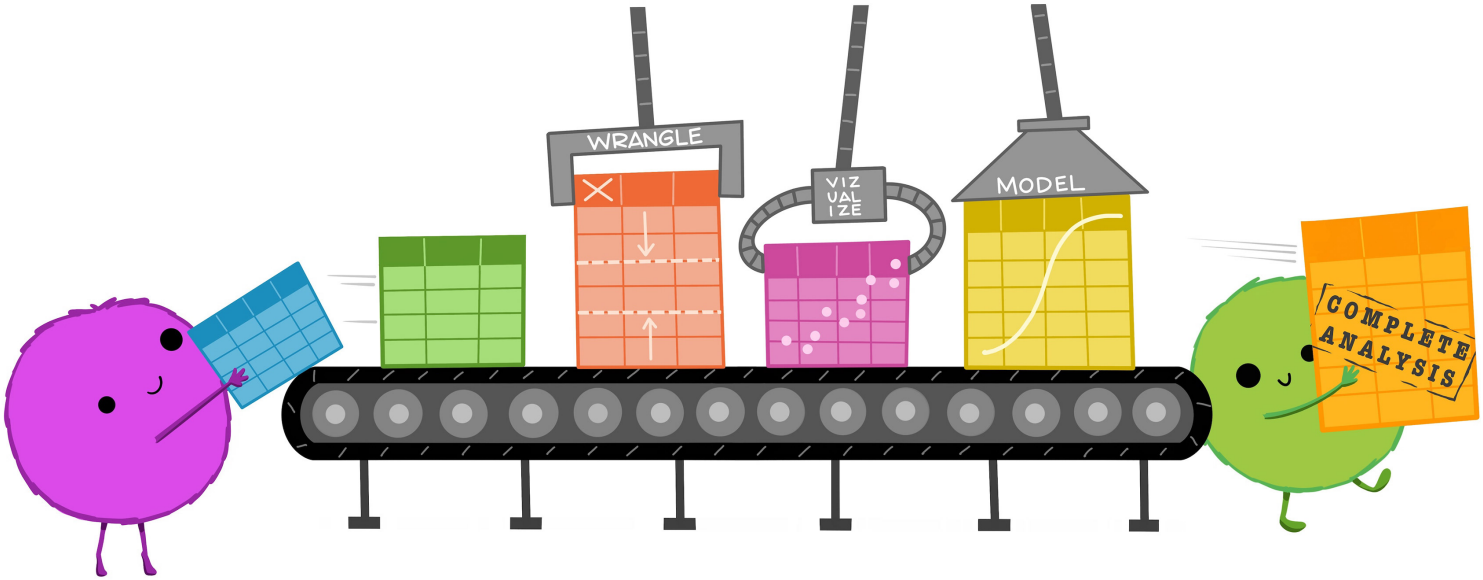
Data Analysis Starts from Data Management

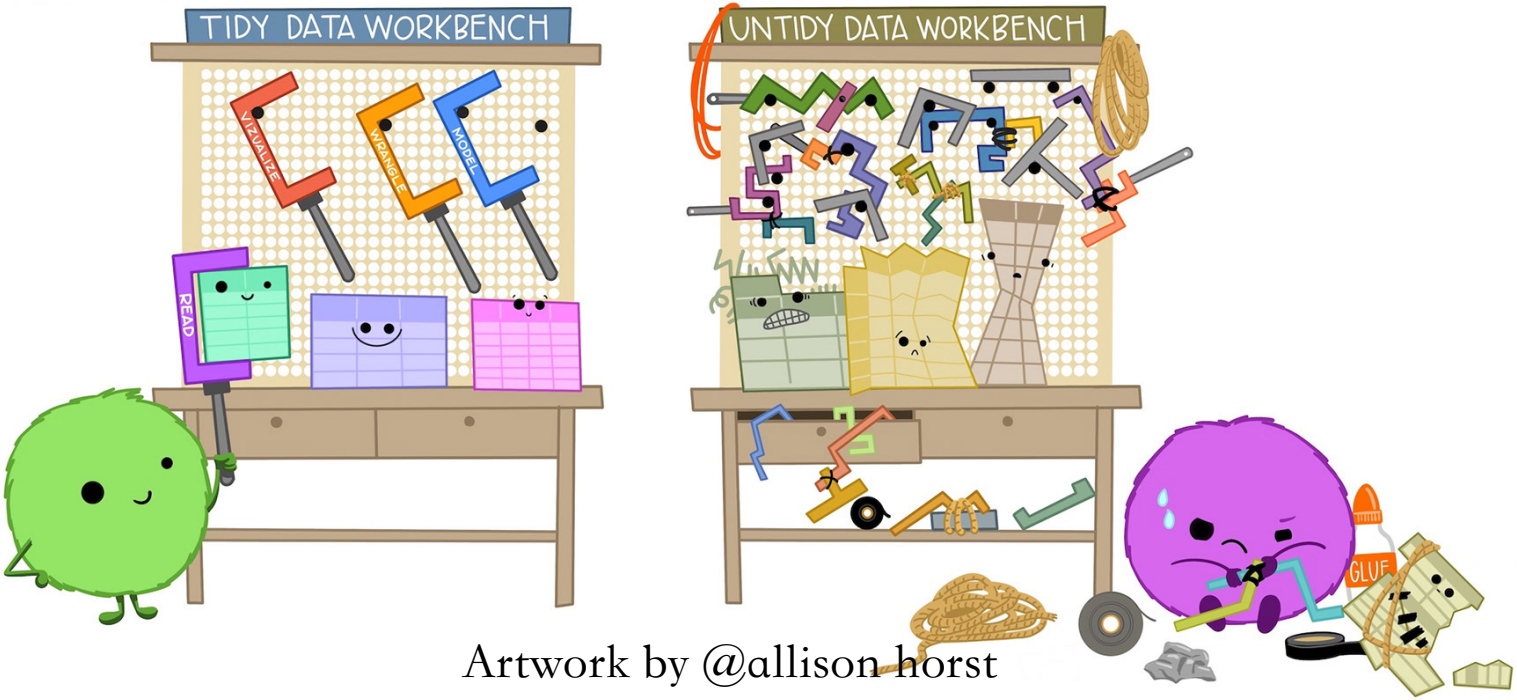


Artwork by @allison horst

Data Analysis Starts from Data Management

When working with tidy data, we can use the same tools in similar ways for different datasets...

...but working with untidy data often means reinventing the wheel with one-time approaches that are hard to iterate or reuse.



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Data Analysis Starts from Data Management

"TIDY DATA is a standard way of mapping the meaning of a dataset to its structure."

–HADLEY WICKHAM

In tidy data:

- each variable forms a column
- each observation forms a row
- each cell is a single measurement

each column a variable

id	name	color
1	floof	gray
2	max	black
3	cat	orange
4	donut	gray
5	merlin	black
6	panda	calico

each row an observation

Wickham, H. (2014). Tidy Data. Journal of Statistical Software *59* (10). DOI: 10.18637/jss.v059.i10

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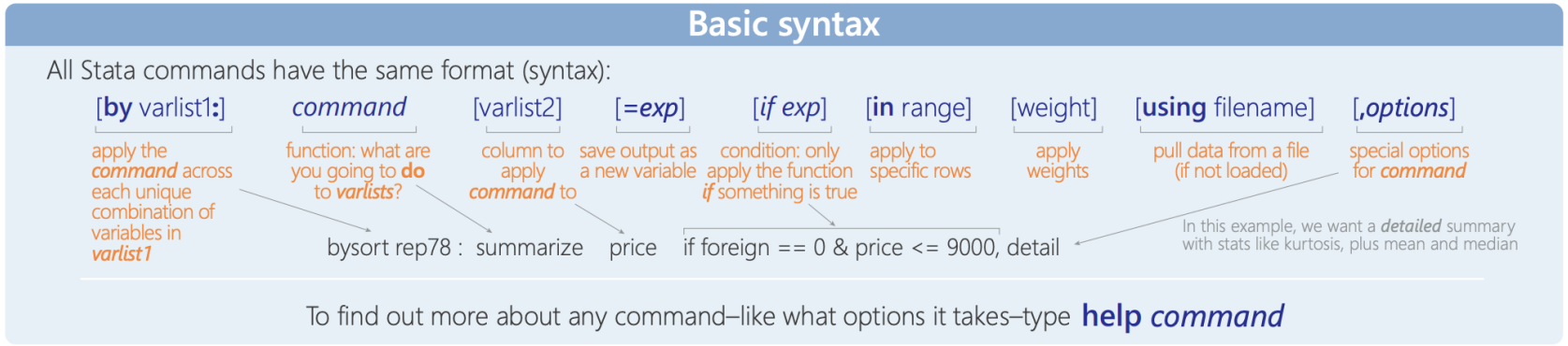
Outline

Basics

Open, Browse, and Export Data

Data Management
- Variable Type, Storage Type, and Formats
- Generate Variables and Change Values
- Labels
- Select, Sort, and Save Data

Basic Syntax



stata cheatsheet

- Knowing the basic syntax is helpful when you read the help document.
- Parts within the brackets are optional.

Basic Operations

Basic data operations

Arithmetic

- `+` add (numbers) combine (strings)
- `-` subtract
- `*` multiply
- `/` divide
- `^` raise to a power

Logic

- `&` and
- `!` or `~` not
- `|` or

`==` tests if something is equal
`=` assigns a value to a variable

- `==` equal
- `!=` or `~=` not equal
- `<` less than
- `<=` less than or equal to
- `>` greater than
- `>=` greater or equal to

if foreign != 1 & price >= 10000

make	foreign	price
Chevy Colt	0	3,984
Buick Riviera	0	10,372
Honda Civic	1	4,499
Volvo 260	1	11,995

if foreign != 1 | price >= 10000

make	foreign	price
Chevy Colt	0	3,984
Buick Riviera	0	10,372
Honda Civic	1	4,499
Volvo 260	1	11,995

stata cheatsheet

- Note the difference between $=$ and $==$.

STATA as Calculator: "display"

```
. * addition
. display 3+6 //for addition calulation
9
. * substraction
. display 109-5.6
103.4
. * multiplication
. dis 11*35
385
. * division
. dis 34/8
4.25
. * power
. dis 23^3
12167
```

STATA as Calculator: Math functions

```
. *** some math functions
. dis sqrt(6)
2.4494897
. dis int(5.2)
5
. dis log(4)
1.3862944
. dis exp(2.39)
10.913494
. dis min(2.1, 3, 7)
2.1
. dis max(2.1, 3, 7)
7
. dis sum(2 3 4 5 6)
23456
. *NOTE: type "help math_functions" to see available math functions
.
. ** more complicated calculations
. dis (min(2.1,3,9)*max(2.1,3,9)/sqrt(2))^2
178.605
. dis exp(1.2)/(exp(1.2)+1)
.76852478
```

- Type "help math_functions" for more.

String Functions

```
. display "Hello there" //Strings should be put in quotation marks: ""
Hello there

.
. dis length("This string has 29 characters")
29

. dis strpos("Stata", "a")
3

. dis substr("Stata", 3, 5)
ata

. dis regexr("My string", "My", "Your")
Your string

. dis trim(" leading / trailing spaces ")
leading / trailing spaces

. dis strlower("STATA should not be ALL-CAPS")
stata should not be all-caps
```

- Find more string functions here
- We can never remember every function, but we can always google and use help.

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Input Data: "input"

We can type in data in STATA using input.

```
. *** Input data
. clear //clear data in memory
. input x y

            x          y
  1.      6 0
  2.      7 .
  3. end
. save "$outer/data/input.dta", replace
file /Users/bing/Dropbox/MGCS5010_2023Spring/1.Lectures/STATATutorial/data/input
> .dta saved
.          // save the input data to .dta form
. * use "replace"
```

"clear, replace"

Clear

- Stata can only open one dataset at once.
- "clear" the data in memory (especially when you've made changes to the data) before open a new one.

Replace

- save and "replace" the already existed file.

Open and Save .dta Data: "use, save"

To open .dta data, simply use use.

```
. *** Read in and save Stata format data
. use "$outer/data/cfps2010adult_10%.dta", clear
.         // open .dta data
. * We can also put "clear" as the option for use.
.         keep pid provcd urban gender
. save "$outer/data/cfps10_part.dta", replace
file /Users/bing/Dropbox/MGCS5010_2023Spring/1.Lectures/STATATutorial/data/cfps1
> 0_part.dta saved
.         // save data with a new name
. * Why don´t we just overwrite the original data ???
. * Use "replace" to replace the already existed file, when we rerun the codes.
. saveold "$outer/data/cfps10_part_old.dta", replace
(saving in Stata 13 format)
(FYI, saveold has options version(12) and version(11) that write files in
      older Stata formats)
file /Users/bing/Dropbox/MGCS5010_2023Spring/1.Lectures/STATATutorial/data/cfps1
> 0_part_old.dta saved
.         //save old version data
```

Open and Export Data of Other Formats: "import, export"

- Use "help import" to explore how it works.

```
. *** Other data formats
. help import
. * Let´s first export current data into excel file
. export excel "$outer/data/cfps10_part.xlsx", first(varl) replace
file /Users/bing/Dropbox/MGCS5010_2023Spring/1.Lectures/STATATutorial/data/cfps1
> 0_part.xlsx saved
.         //save current data to excel
. import excel "$outer/data/cfps10_part.xlsx", first clear
.         //open excel format data
. * Note that after such transformation, the variable pid has some problem
.         * 1. It is due to the data format: string vs number, and the lenghth o
> f the variable.
.         * 2. Be careful when we use other forms of data.
```

Exercise

- Download the "cfps10_foruse.dta", and save it to a proper place.
- Write codes in your do file to open the data.

Explore Data

When we read the data into memory, we can explore it quickly before we continue working on with it.

- browse
- describe
- codebook
- count
- list

```
. describe age gender

              storage   display    value
variable name   type    format     label      variable label
------------------------------------------------------------------------------
age             float   %9.0g                 Age in Survey Year
gender          float   %9.0g      gender     Gender

. codebook han

------------------------------------------------------------------------------
han                                                                       Han
------------------------------------------------------------------------------

                  type:  numeric (float)
                 label:  han

                 range:  [0,1]                        units:  1
         unique values:  2                        missing .:  13/3,360

            tabulation:  Freq.   Numeric  Label
                           256         0  Ethnic Minorities
                         3,091         1  Han
                            13         .

. count // count the number of observations
  3,360

. count if age==20 // count N whose age equals 20
  55
```

```
. list pid if age==94
```

	pid
2331.	441676102

```
. list pid in 1/10
```

	pid
1.	110011101
2.	110013101
3.	110020103
4.	110030101
5.	110035103
6.	110044104
7.	110050101
8.	110052103
9.	110065102
10.	110066102

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Data Management

Now we can make changes to the data and get the tidy and simple data we need. Common operations include:

- convert variable types
- rename variable names
- generate new variables
- add labels
- replace and recode values
- select rows and columns
- sort and order
- ...

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Variable Type, Storage Type, and Formats

Variable Type

- Numeric
- String

Storage types

- For numeric: byte, int, long, float, double
- For string: str1, str2, ..., strL
- help data_types

Formats

- the data display and output format, for example:
  1. digits after decimal point
  2. leading zeros
- help format

Convert Variable Types: "to(de)string, en(de)code"

```
. * convert numeric variables to string variables: tostring, decode
. tostring gender, gen(Gender)
Gender generated as str1

. decode gender, gen(GENDER)

. describe gender Gender GENDER

              storage   display    value
variable name   type    format     label      variable label
-------------------------------------------------------------------------------
gender          float   %9.0g      gender     Gender
Gender          str1    %9s                   Gender
GENDER          str6    %9s                   Gender

. * compare between tostring and decode
. * tabulate categorical variables
. tab Gender // tostring convert the "values" of a numeric variable

    Gender |      Freq.     Percent        Cum.
-----------+-----------------------------------
         0 |      1,770       52.68       52.68
         1 |      1,590       47.32      100.00
-----------+-----------------------------------
     Total |      3,360      100.00

. tab GENDER // decode convert the "labels" of a numeric variable

    Gender |      Freq.     Percent        Cum.
-----------+-----------------------------------
    Female |      1,770       52.68       52.68
      Male |      1,590       47.32      100.00
-----------+-----------------------------------
     Total |      3,360      100.00
```

Convert Variable Types: "to(de)string, en(de)code"

```
. * convert string variables to numeric variables: destring, encode
. destring GENDER, gen(gender1) //destring cannot work with string variab
> le with "words"
GENDER: contains nonnumeric characters; no generate

. destring Gender, gen(gender2)
Gender: all characters numeric; gender2 generated as byte

. encode GENDER, gen(gender3) //encode generate numeric variable with val
> ues

. encode Gender, gen(gender4) //again, encode generate numeric variable w
> ith values

. tab gender3
```

Gender	Freq.	Percent	Cum.
Female	1,770	52.68	52.68
Male	1,590	47.32	100.00
Total	3,360	100.00	

```
. tab gender3, nolabel // tab without label
```

Gender	Freq.	Percent	Cum.
1	1,770	52.68	52.68
2	1,590	47.32	100.00
Total	3,360	100.00	

```
. label list gender3
gender3:
           1 Female
           2 Male

. label list gender4
gender4:
           1 0
           2 1
```

Storage type: "recast"

```
. des pid

              storage   display    value
variable name   type    format     label      variable label
------------------------------------------------------------------------------
pid             double  %12.0g     pid        Individual ID

. recast float pid
pid:  3271 values would be changed; not changed

. des pid

              storage   display    value
variable name   type    format     label      variable label
------------------------------------------------------------------------------
pid             double  %12.0g     pid        Individual ID
```

Data Formats: "format"

```
. format %9.0g pid //change the format for pid
. format pid //check the format for pid
  variable name  display format
  ----------------------------------
  pid            %9.0g
  ----------------------------------

. format %12.0g pid
. format pid
  variable name  display format
  ----------------------------------
  pid            %12.0g
  ----------------------------------
```

Data Formats: "format"

```
. list rswt_nat in 1/5

     +-----------+
     |  rswt_nat |
     |-----------|
  1. | 37606.55469 |
  2. | 43263.13281 |
  3. | 55504.42578 |
  4. | 29866.04297 |
  5. | 46588.98828 |
     +-----------+

. des rswt_nat

                storage   display    value
variable name     type    format     label      variable label
-------------------------------------------------------------------------------
rswt_nat        double  %12.0g     rswt_nat   Weight_National

. format %9.2f rswt_nat

. list rswt_nat in 1/5

     +----------+
     | rswt_nat |
     |----------|
  1. | 37606.55 |
  2. | 43263.13 |
  3. | 55504.43 |
  4. | 29866.04 |
  5. | 46588.99 |
     +----------+

. des rswt_nat

                storage   display    value
variable name     type    format     label      variable label
-------------------------------------------------------------------------------
rswt_nat        double  %9.2f      rswt_nat   Weight_National
```

Variable Name: "rename"

```
. * rename a single variable
. rename gender sex
. * help rename group
. rename (gender2 gender3 gender4) (sex2 sex3 sex4)
. rename sex* gender*
. drop GENDER
. rename gender*, upper
. rename GENDER, lower
```

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Generate New Variable: "gen, egen"

```
. generate birthy = 2010 - age
. egen meanage = mean(age)
. tab meanage
```

meanage	Freq.	Percent	Cum.
45.56964	3,360	100.00	100.00
Total	3,360	100.00	

```
. bysort gender: egen meaneduy = mean(eduy)
. tab meaneduy
```

meaneduy	Freq.	Percent	Cum.
6.145198	1,770	52.68	52.68
8.058491	1,590	47.32	100.00
Total	3,360	100.00	

Change Variable Values: "replace"

```
. gen educated = . //generate a numeric variable with missing values .
(3,360 missing values generated)

. replace educated = 0 if eduy==0
(804 real changes made)

. replace educated = 1 if eduy>0 & eduy!=.
(2,556 real changes made)

. tab educated

   educated |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |        804       23.93       23.93
          1 |      2,556       76.07      100.00
------------+-----------------------------------
      Total |      3,360      100.00

.
. gen EDUCATED = "" //generate a string variable with missing values ""
(3,360 missing values generated)

. replace EDUCATED = "Not Educated" if eduy==0
variable EDUCATED was str1 now str12
(804 real changes made)

. replace EDUCATED = "Educated" if eduy>0 & eduy!=.
(2,556 real changes made)

. tab EDUCATED

    EDUCATED |      Freq.     Percent        Cum.
-------------+-----------------------------------
    Educated |      2,556       76.07       76.07
Not Educated |        804       23.93      100.00
-------------+-----------------------------------
       Total |      3,360      100.00
```

Change Variable Values: "recode"

```
. recode age (16/25=1) (26/35=2) (36/45=3) (46/55=4) (56/65=5) (else=.), gen(age_cu
(3360 differences between age and age_cut)

. tab age_cut
```

RECODE of age (Age in Survey Year)	Freq.	Percent	Cum.
1	475	16.04	16.04
2	455	15.36	31.40
3	798	26.94	58.34
4	659	22.25	80.59
5	575	19.41	100.00
Total	2,962	100.00	

Missing Values: "mvdecode"

- Missing values
  - . for numeric variables
  - "" for string variables
- We can use replace or recode to change some values into missing.
- We can also use "mvdecode"

```
. count if income==.
  0
. count if income<0
  872
. mvdecode income, mv(-9 10)
      income: 724 missing values generated
. count if income==.
  724
```

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Label Variables, Values, and Data: "label"

```
. label variable educated "Ever received any education"
. label variable age_cut "Age Categories"
. des educated age_cut

              storage   display    value
variable name   type    format     label      variable label
-------------------------------------------------------------------------------
educated        float   %9.0g                 Ever received any education
age_cut         float   %9.0g                 Age Categories
```

Label Variables, Values, and Data: "label"

```
. label define yesno 1 "Yes" 2 "No" //define value label
. label value educated yesno //add value label to categorical variables
. tab educated

      Ever |
  received |
       any |
 education |      Freq.     Percent        Cum.
-----------+-----------------------------------
         0 |        804       23.93       23.93
       Yes |      2,556       76.07      100.00
-----------+-----------------------------------
     Total |      3,360      100.00

. tab educated, nolab

      Ever |
  received |
       any |
 education |      Freq.     Percent        Cum.
-----------+-----------------------------------
         0 |        804       23.93       23.93
         1 |      2,556       76.07      100.00
-----------+-----------------------------------
     Total |      3,360      100.00

. des educated

              storage   display    value
variable name   type    format     label      variable label
-------------------------------------------------------------------------------
educated        float   %9.0g      yesno      Ever received any education
```

Label Variables, Values, and Data: "label"

```
. label data "CFPS 2010 for STATS Tutorial"

. describe, short

Contains data from /Users/bing/Dropbox/MGCS5010_2023Spring/1.Lectures/STATATutorial/data/cfps
> 10_foruse.dta
  obs:         3,360                          CFPS 2010 for STATS Tutorial
 vars:            45                          4 Sep 2022 23:35
 size:       705,600
Sorted by: gender
     Note: Dataset has changed since last saved.
```

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Select Rows and Columns: "keep, drop"

```
. * keep and drop rows
. keep if income!=.
(724 observations deleted)

. drop if age>65
(257 observations deleted)

.
. * keep and drop variables
. drop Gender GENDER2

. drop birthy-age_cut

. keep pid-ccp
```

Sort Rows and Order Columns: "(g)sort, order"

```
. sort age
. gsort -age
. gsort -age eduy
.
. order pid fid urban hukou han
. order rswt_nat, last
```

Save Modified Data

- Remember to save the modified data if needed.
- You can first "compress" before saving.

```
. compress //compress to proper storage type to save storage
  variable urban was float now byte
  variable han was float now byte
  variable age was float now byte
  variable gender was float now byte
  variable edu was float now byte
  variable eduy was float now byte
  variable employ was float now byte
  variable employs was float now byte
  variable occtype_crude was float now byte
  variable occisei was float now byte
  variable tenure was float now byte
  variable nsib was float now byte
  variable fedu was float now byte
  variable medu was float now byte
  variable focctype_isco was float now int
  variable mocctype_isco was float now int
  variable foccisei was float now byte
  variable moccisei was float now byte
  variable word was float now byte
  variable math was float now byte
  variable marriage was float now byte
  variable sbirthy was float now int
  variable sedu was float now byte
```